

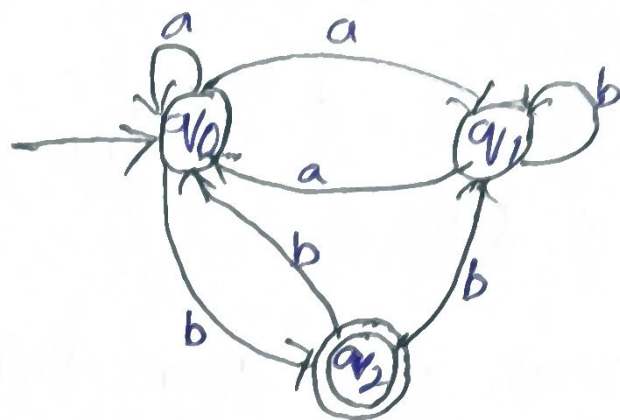
CO1 - Assessment - 01

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1) construction of DFA from the given NFA

1) NFA Transition table

state	a	b
q_0	$\{q_0, q_1\}$	$\{q_2\}$
q_1	$\{q_0\}$	$\{q_1\}$
q_2	$\emptyset$	$\{q_0, q_1\}$



2) Substrat construction

start state = $\{q_0\}$

$$\begin{aligned} \delta(\{q_0\}, a) &= \{q_0, q_1\} \\ \delta(\{q_0\}, b) &= \{q_2\} \end{aligned} \quad \left. \vphantom{\begin{aligned} \delta(\{q_0\}, a) &= \{q_0, q_1\} \\ \delta(\{q_0\}, b) &= \{q_2\} \end{aligned}} \right\} \text{from } \{q_0\}$$

$$\begin{aligned} \delta(\{q_0, q_1\}, a) &= \{q_0, q_1\} \\ \delta(\{q_0, q_1\}, b) &= \{q_1, q_2\} \end{aligned} \quad \left. \vphantom{\begin{aligned} \delta(\{q_0, q_1\}, a) &= \{q_0, q_1\} \\ \delta(\{q_0, q_1\}, b) &= \{q_1, q_2\} \end{aligned}} \right\} \text{from } \{q_0, q_1\}$$

$$\begin{aligned} \delta(\{q_2\}, a) &= \emptyset \\ \delta(\{q_2\}, b) &= \{q_0, q_1\} \end{aligned} \quad \left. \vphantom{\begin{aligned} \delta(\{q_2\}, a) &= \emptyset \\ \delta(\{q_2\}, b) &= \{q_0, q_1\} \end{aligned}} \right\} \text{from } \{q_2\}$$

$$\left. \begin{aligned} \delta(\{q_1, q_2\}, a) &= \{q_0\} \\ \delta(\{q_1, q_2\}, b) &= \{q_0, q_1\} \end{aligned} \right\} \text{from } (q_1, q_2)$$

$$\left. \begin{aligned} \delta(\emptyset, a) &= \emptyset \\ \delta(\emptyset, b) &= \emptyset \end{aligned} \right\} \text{from } \emptyset$$

3. Final DFA transition table

State	a	b
$A = \{q_0\}$	B	C
$B = \{q_0, q_1\}$	B	D
$C = \{q_2\}$	E	B
$D = \{q_1, q_2\}$	A	B
$E = \emptyset$	E	E

$$A = \{q_0\}$$

$$B = \{q_0, q_1\}$$

$$C = \{q_2\}$$

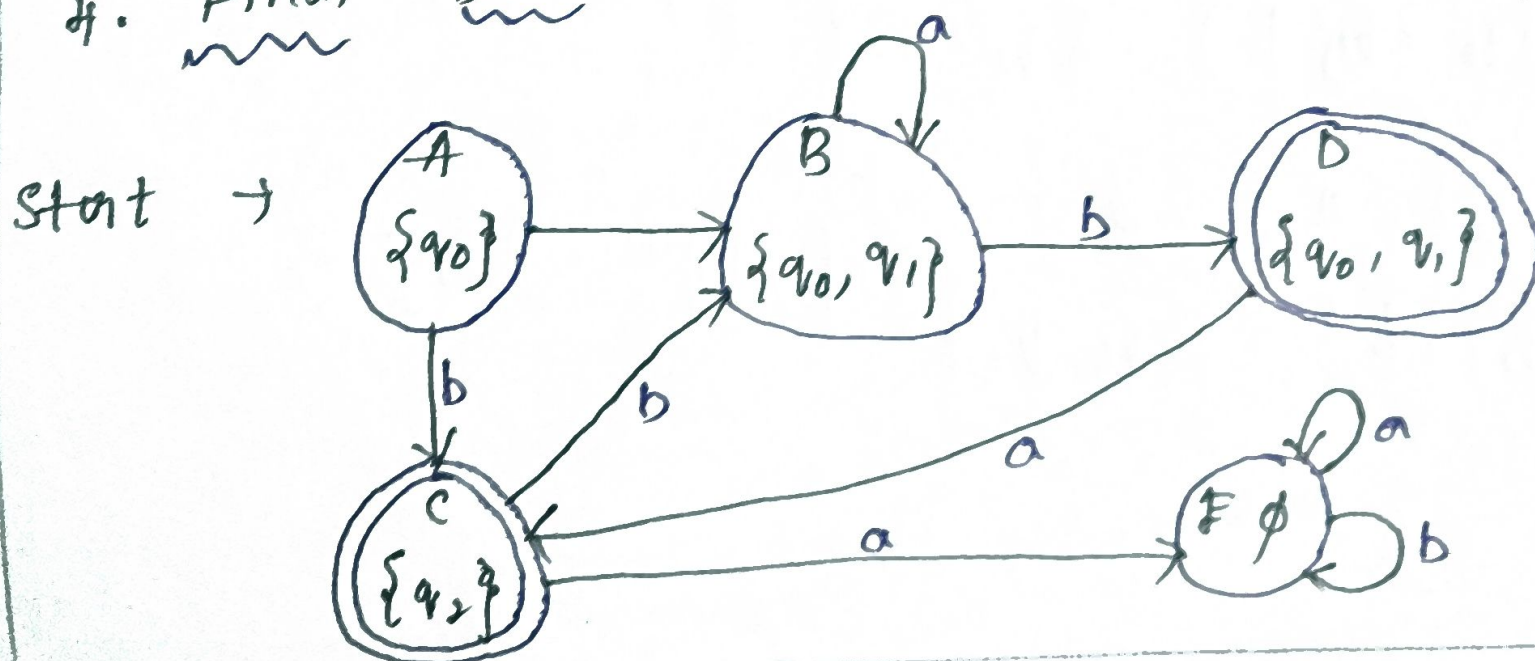
$$D = \{q_1, q_2\}$$

$$E = \emptyset \text{ (dead state)}$$

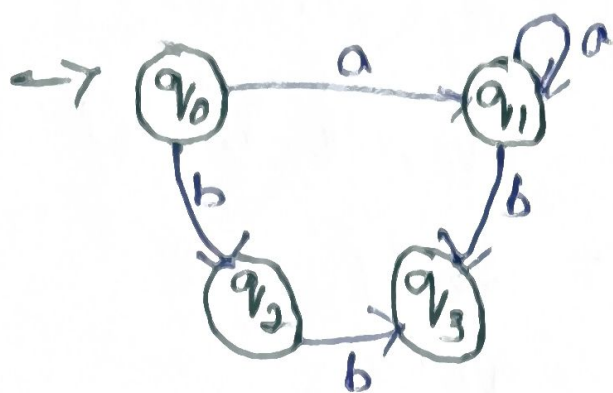
Start State = A

Final state = C, D

4. Final DFA



Alphabet $\Sigma = \{a, b\}$



NFA Transition table

State	a	b
q_0	$\{q_1\}$	$\{q_2\}$
q_1	$\{q_1\}$	$\{q_3\}$
q_2	$\{q_3\}$	$\{q_2\}$
q_3	$\emptyset$	$\emptyset$

Equivalent DFA

DFA Transition table

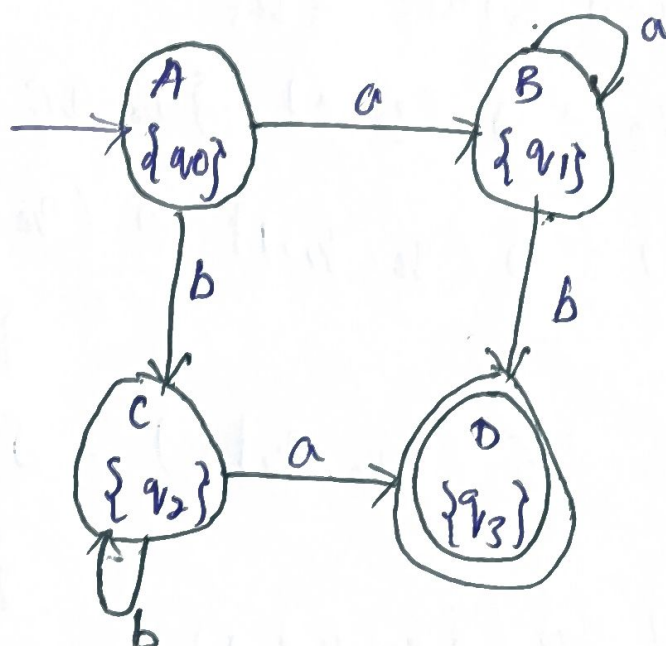
DFA State a b

A = $\{q_0\}$ B C

B = $\{q_1\}$ B D

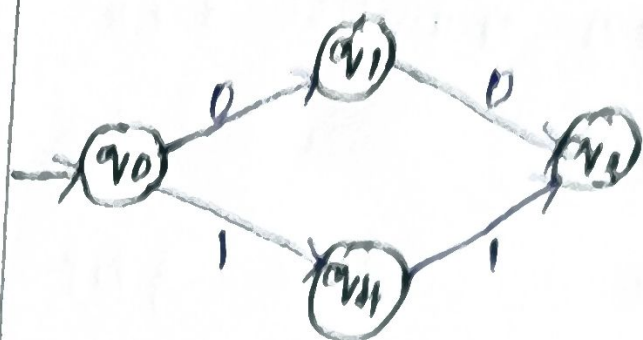
C = $\{q_2\}$ D C

D = $\{q_3\}$ $\emptyset$ $\emptyset$



All unspecified transitions go to dead state $\emptyset$
 ($\emptyset$ has self-loops on both a and b)

3) NFA state diagram



Formal definition of the NFA

$$M = \{Q, \Sigma, \delta, q_0, F\}$$

$$Q = \{q_0, q_1, q_2, q_3\}$$

$$\Sigma = \{0, 1\}$$

$$q_0 = q_0$$

$$F = \{q_3\}$$

$$\delta \quad 0 \quad 1$$

$$q_0 \quad \delta(q_0, 0) = \delta(q_0, q_1)$$

$$q_1 \quad \delta(q_1, 0) = \delta(q_3, q_3)$$

$$q_2 \quad \emptyset \quad \emptyset$$

$$q_3 \quad \emptyset \quad \emptyset$$

String Acceptance Verification: 01011

Initial state $\{q_0\}$

$$\text{Read 0} : \delta(q_0, 0) = \{q_0, q_1\}$$

$$\begin{aligned} \text{Read 1} : \delta(q_0, q_1, 1) &= \delta(q_0, 1) \cup \delta(q_1, 1) \\ &= \{q_0, q_2\} \cup \emptyset = \{q_0, q_2\} \end{aligned}$$

$$\begin{aligned} \text{Read 0} : \delta(\{q_0, q_2\}, 0) &= \delta(q_0, 0) \cup \delta(q_2, 0) \\ &= \{q_0, q_1\} \cup \emptyset = \{q_0, q_1\} \end{aligned}$$

$$\begin{aligned} \text{Read 1} : \delta(\{q_0, q_1\}, 1) &= \delta(q_0, 1) \cup \delta(q_1, 1) \\ &= \{q_0, q_2\} \cup \emptyset = \{q_0, q_2\} \end{aligned}$$

$\therefore$ So, 01011 is accepted
The path accepting: $q_0 \rightarrow q_0 \rightarrow q_0 \rightarrow q_0 \rightarrow q_2 \rightarrow q_3$

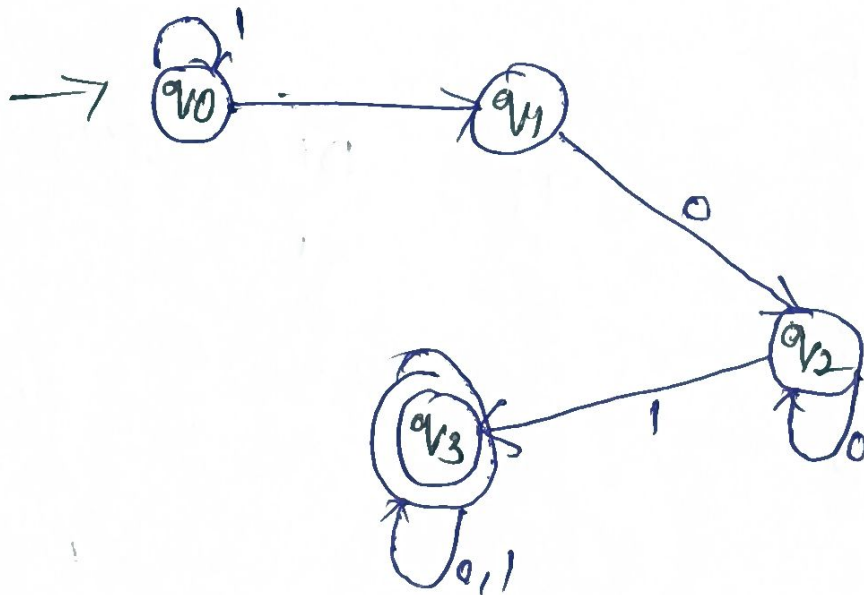
Equivalent DFA

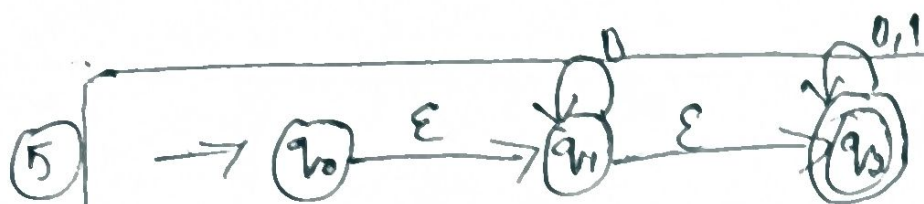
$q_0 \rightarrow \text{start}$ $q_1 \rightarrow \text{method}(0)$ $q_2 \rightarrow (00)$ $q_3 \rightarrow (001)$

Transition table

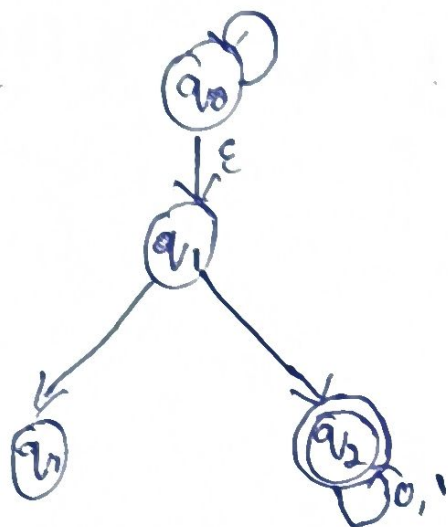
state	0	1
q_0	q_1	q_0
q_1	q_2	q_0
q_2	q_2	q_3
q_3	q_3	q_3

DFA Diagram





Ans:-



Start state = q_0
Final State = q_1

Find enclosure: (q_0)

DFA state are:

$A = \{q_0, q_1, q_2\}$

$B = \{q_1, q_2\}$

$C = \{q_2\}$

① $q_0 \rightarrow \epsilon$ $q_1 \rightarrow \epsilon$

ϵ -closure (q_0) = (q_0, q_1, q_2)

Since each contain q_2
→ A, B, C are final states

③ ϵ -closure (q_2)

No, ϵ -transition! so!

ϵ -closure (q_2) = q_2

DFA transition table:

DFA States

$A = (q_0, q_1, q_2)$

$B = (q_1, q_2)$

$C = q_2$

0:

B

B

C

1:

A

C

C

DFA diagram

